

SignJoy Desktop App

Setup & Compilation Guide

This guide details the step-by-step instructions for setting up the **SignJoy** desktop application in development mode, downloading required offline acoustic models, compiling it to a standalone binary folder, and wrapping it into a single-file executable setup installer (.exe) for distribution.

Prerequisites

Make sure the target machine has the following installed:

- **Windows OS** (Windows 10 or 11)
- **Python 3.10 to 3.12** (with Pip installed and added to the environment variables)
- **Git** (to clone the source repository)
- **Webcam & Microphone** (required for gesture tracking and offline speech recognition)

Step 1: Clone the Repository

Open PowerShell or Command Prompt and clone the code repository from GitHub:

```
git clone https://github.com/teamcore38-droid/Sign-Joy-Unity.git
cd Sign-Joy-Unity
```

Step 2: Set Up the Python Virtual Environment

Navigate to the main AI project directory, create a Python virtual environment, and install all required runtime and desktop packaging packages:

```
cd "-Sign-Joy_AI"
python -m venv .venv
.venv\Scripts\Activate.ps1
python -m pip install --upgrade pip
pip install -r requirements.txt
pip install pywebview vosk pyaudio
```

Step 3: Run the Application in Development Mode

You can run the application immediately to verify the code and local UI before compiling it:

```
python ..\SignJoyDesktop\desktop_launcher.py
```

Step 4: Download & Configure the Offline Speech Model

Run the automated speech recognition model configuration script to download and unpack the lightweight English Vosk acoustic model (~40MB) into the correct directory:

```
python ..\SignJoyDesktop\download_vosk_model.py
```

Step 5: Compile to an Executable Folder

Use PyInstaller to compile Python code, layouts, static web assets, offline translators, and the Vosk speech model into a single binary folder structure:

```
python ..\SignJoyDesktop\build_desktop_app.py
# The output will be located in the folder: dist\SignJoy\
```

Step 6: Build the Standalone Setup Installer (.exe)

To package the compiled directory into a professional, double-clickable installer wizard: 1. Download and install **Inno Setup 6** (<https://jrsoftware.org/isdl.php>). 2. Launch Inno Setup and open:

SignJoyDesktop\installer_config.iss 3. Select **Build > Compile** (or press **F9**) to generate the installer. 4. The single setup file **SignJoy_Offline_Setup.exe** will be built at: **SignJoyDesktop\dist\setup**